FEDERAL INSTITUTE OF SCIENCE AND TECHNOLOGY (FISAT)

Hormis Nagar, Mookkannoor PO, Angamaly, Kochi
Accredited by NAAC with 'A+' Grade

DEPARTMENT OF COMPUTER APPLICATIONS

SYNOPSIS OF THE MINI PROJECT

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Project Title	AI-Powered Knowledge Base Chatbot
Area of the Project	AI/ Natural Language Processing (NLP) / Generative AI
Date of Submission	20-07-2026

Intelligent Knowledge Base and Information Retrieval System Using RAG

This project is a full-stack AI-powered Knowledge Base Chatbot built with React.js, Node.js, Express, MongoDB, and Google Gemini. The system allows administrators to upload knowledge documents such as PDF, PPTX, and TXT files, which are extracted and divided into smaller chunks for efficient processing. These chunks are converted into vector embeddings using the Google Gemini embedding model and stored in MongoDB. When users ask questions, the system performs semantic search using MongoDB Atlas Vector Search to find the most relevant information from the uploaded documents. The retrieved content is then provided to Google Gemini to generate accurate and context-based answers using the Retrieval-Augmented Generation (RAG) approach. The system also maintains chat history and displays the relevant document sources used for generating responses. In addition, the Daily Highlights feature allows administrators to publish important updates, announcements, and information for users in a simple and easy-to-understand format.

Front End & Back End Tools	**Front End:** React.js, Vite, CSS **Back End:** Node.js, Express.js, Google Gemini API **Database:** MongoDB Atlas